

Could you please inform about Team Avishkar Hyperloop and how GalaxEye was founded?

We have emerged from the Avishkar Hyperloop team at IIT Madras. When I started my master's in aerospace engineering at IIT Madras in 2017, I got to know about the opportunity at SpaceX, where Elon Musk was organising this competition called 'Hyperloop Contest.'

I got an opportunity to form a team and we worked for a couple of years to build the hyperloop pod, which can go from one place to another inside a vacuum tunnel at a very high speed. So that was the concept, and a lot of students gathered together, and we were able to do well in the competition. In fact, we were the only Asian team to participate, reach the finals, and finish among the top ten.

A few of us from the hyperloop team were excited about space technologies in general. When we were at the SpaceX facility, we met a lot of folks—SpaceX employees as well as Elon Musk. That sparked our interest, and we thought possibly we could innovate in space tech as well.

We are from different backgrounds—aerospace, engineering design, and metallurgy. We came together and figured out a sweet spot: building satellites. Satellites were our interest because they include hardware and output data, which is useful for analytics. And that's how our journey started—by figuring out the purpose of our satellites and how we could differentiate ourselves.

Having a mentor is important in a niche field like space tech. Would you like to talk about your mentors or anyone in particular who has helped you in your journey so far?

We are working on space electronics, and there are multiple categories of electronics involved.

If I have to make something work right now, on the table, I can pretty much use any electronics. When it has to work on a drone, or has to have repeatability, I have to use industrial electronics. If it's on an aircraft, I'll have to use military-grade electronics, and then space-grade.

So, it is very important for us to understand the nature of electronics that can be used in space. That's why we have mentors broadly divided into two categories—people who have an aerospace background

DRISHTI SENSORS ARE A VERY, VERY KEY TECHNOLOGY AND ALSO THE USP OF GALAXEYE.

and people who have an electronics background. Our mentors from the aerospace background tell us about what kind of environment the electronics will be exposed to, and people from electronics tell us how to make the sensors and the satellite function.

Professor Satya, who is from the Department of Aerospace Engineering at IIT Madras, helps us with the aerospace engineering side. And then there are a few senior advisors whom I can't name as of now, but they are from the electronics side and have been associated with a few reputed government organisations earlier.

Are there any particular open source resources that you had used and would like to mention?

We referred to very little open source material, but there are some good repositories from ISRO and NASA that we found online. For most things, we have to get proprietary information. But there are some

very good books written in the past by scientists that we use consistently to build upon our hypotheses and experiment further.

What is the key technology you have implemented? Could you elaborate on Drishti sensors and how your team came up with them?

There is a bigger machine in place—the satellite—which is where the computing also happens. But the Drishti sensors are a very, very key technology and also the USP of GalaxEye. There are multiple sensors today that are used on individual basis. For example, you can use cameras, thermal sensors, radiometers, or radars on a satellite. Drishti sensors are basically a hybrid version of multiple sensors that enable data fusion to give comprehensive data that anyone would want from a satellite. We have filed for a patent for the Drishti sensor.

Initially, we talked to a lot of people to understand their requirements. We figured out the kind of data that was needed; the sensor that we've developed suits those kinds of needs. The sensors themselves are not commercially available; these will be deployed on our satellites only. What we will be making commercially available is the dataset obtained.

Which sectors in the market will the data help?

We're combining multiple datasets together. As I said, data fusion can solve many problems in the industry. It can serve insurance, agriculture, utilities, defense, and many other sectors. Given the comprehensive nature of our dataset, it can serve almost every sector that uses satellite imagery as a data source.

Are there any customer-related challenges that you have identified?

I think it's the volume of the data that will be transferred from